

- This exam has 5 questions for a total of 100 marks.
- Warning: CMI's academic policy regarding cheating applies to this exam.
- You can use results seen in class without reproving them.
- Keep your answers precise. A short, clear, precise answer is always preferred over a long, ambiguous answer.
- "Game" always refers to a strategic form game.

1. Show that mixed Nash equilibrium computation in three-player zero-sum games is as hard as general two-player games (and hence PPAD-complete). *Hint: Show that any two-player game can be converted to a three-player zero-sum game.* [15]

2. Consider the fair division of indivisible items, when the items are *chores* i.e. they are undesirable. There are  $n$  agents,  $m$  indivisible items to be distributed among the agents, each agent  $i$  has a value  $v_i(g) \leq 0$  for an item  $g$ . Value that agent  $i$  assigns to a subset of items  $S$  is  $\sum_{g \in S} v_i(g)$ . An allocation  $(X_1, \dots, X_n)$  is said to be EF1 if,  $\forall i \ \forall j \ \exists g \in X_i$  such that  $v_i(X_i \setminus \{g\}) \geq v_i(X_j)$ .<sup>1</sup>

(a) Consider an envy-cycle with respect to an EF1 allocation  $X = (X_1, \dots, X_n)$ : agent  $i$  envies  $j$ ,  $j$  envies  $k$  and  $k$  envies  $i$ . We have seen in class that, in case of  $v_i(g) \geq 0 \ \forall g$ , a cyclic transfer of bundles where  $i$  gets  $X_j$ ,  $j$  gets  $X_k$ ,  $k$  gets  $X_i$  preserves the EF1 property. Give an example to show that a cyclic transfer *does not* necessarily preserve the EF1 property when items are chores. *Hint: You may consider another agent  $\ell$  that is not part of the envy-cycle.* [7]

(b) Now consider a modified envy-graph where an edge  $(i, j)$  is present only if  $j = \operatorname{argmax}_{k \in [n]} v_i(X_k)$  and  $v_i(X_j) > v_i(X_i)$ . Show that a cyclic transfer, as described above, in this modified graph preserves the EF1 property. [10]

(c) Using the above observation, conclude that an EF1 allocation for chores can be computed in polynomial time. [10]

3. Consider the following game: A factory is creating pollution which is adversely affecting  $n$  people. Each person has two options: lodging a complaint or keeping quiet. If everyone keeps quiet, the payoff is 0 to everyone. If someone lodges a complaint, everyone receives a payoff of  $v > 0$ , except the person who lodges a complaint loses a payoff  $0 < c < v$  due to the efforts of lodging a complaint. Thus a person who lodges a complaint receives a payoff  $v - c$ .

<sup>1</sup>Note that, in case of positive values, agent  $i$  removed an item from agent  $j$ 's bundle whereas here,  $i$  removes an item from her own bundle. Also, all items must be allocated.



- (a) List all the pure Nash equilibria of this game. *one complaint* [10]  
 (b) Find a symmetric mixed strategy Nash equilibrium. Symmetric means every player follows the same mixed strategy.  *$C = 1 - (\frac{1}{2})^n$*  [10]  
 (c) What is the probability that at least one person lodges a complaint? [10]

4. Consider the following two-player game:

|   | A    | B   | C    |
|---|------|-----|------|
| A | 4,4  | 0,5 | -1,0 |
| B | 5,0  | 1,1 | 0,0  |
| C | 0,-1 | 0,0 | 1,1  |

- (a) Find all pure strategy Nash equilibria.  *$\langle B, B \rangle, \langle C, C \rangle$*  [8]  
 (b) Is there a mixed strategy Nash equilibrium with A in its support for both players? [5]  
 If yes, exhibit one or justify why not. *No*  
 (c) Find all mixed strategy Nash equilibria.  *$\langle 0, \frac{1}{2}, \frac{1}{2} \rangle, \langle 0, \frac{1}{2}, \frac{1}{2} \rangle$*  [8]  
 5. Are the following statements true or false? Justify. [7]

- If  $(i, j)$  and  $(h, k)$  are pure strategy Nash equilibria in a two-player game, then  $(i, k)$  and  $(h, j)$  are also pure strategy Nash equilibria. *coordinating game*
- If  $(i, j)$  and  $(h, k)$  are pure strategy Nash equilibria in a two-player game, then any convex combination of them is a mixed strategy Nash equilibrium. *False*
- If  $(i, j)$  and  $(h, k)$  are the only pure strategy Nash equilibria in a two-player game, then no other strategy apart from  $i, j, h, k$  can be in the support of a mixed strategy Nash equilibrium. *F*

$$\begin{aligned}
 & 3P + 2Q + 1 - P \cdot Q \\
 &= 2P + 3Q + 2 - 2P \cdot Q \\
 &= P + 2Q + 3 - 3P - 3Q
 \end{aligned}$$

$$\begin{aligned}
 & 2P + Q + 1 \\
 &= Q + 2 \\
 &= 3 - P - Q
 \end{aligned}$$

$$P = \frac{1}{2}$$

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$$3P + 2Q + 1 - P \cdot Q$$

$$= 4P + 2Q + 2 - 2P - Q$$

$$= P + 2Q + 3 - 3P - 3Q$$

$$\begin{aligned}
 2P + Q + 1 &= 2P + 2 \\
 &= 3 - 2P - Q
 \end{aligned}$$